

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 30%

QUESTIONS: 2 Total Marks:30 Annexure A

Industry Problem Solving Task

Code of Conduct:

I _Shaik Mahaboob Basha (192365045)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shaik Mahaboob Basha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Industry Problem	20%	Comprehensive understanding of real world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	

Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Annexure A

Industry Problem Solving Task

1: Automated Defect Detection in Manufacturing

A precision engineering company wants to automatically detect defects in mechanical components using computer vision. Parts vary slightly in shape due to manufacturing tolerances.

Questions:

1. How can PCA-based shape priors and contour-based representations be used to model standard component shapes?
2. Propose a pipeline using machine learning (SVM, GMM) or deep learning (CNN) to classify defective vs. non-defective parts.
3. Discuss methods to handle noise, partial occlusions, and varying illumination in industrial images.

2: Autonomous Vehicle Obstacle Recognition

A self-driving car company needs to detect rare or unexpected obstacles on urban roads. The labeled data for these obstacles is limited.

Questions:

1. Explain how data augmentation using GANs can help generate synthetic images for training.
2. Compare the effectiveness of CNNs versus Vision Transformers for detecting small or sparse objects in high-resolution images.
3. Suggest a real-time inference pipeline for obstacle detection, considering computational constraints in embedded systems.

Question 1: Automated Defect Detection in Manufacturing

Problem Understanding

A precision engineering company wants to automatically identify defective mechanical components. Parts vary slightly in shape because of manufacturing tolerances. Key challenges include small shape variations, noise, illumination changes and partial occlusions.

1. PCA-Based Shape Priors and Contour-Based Representations

PCA can learn normal shape variations from non-defective components. The pipeline is:

Image Acquisition → Segmentation → Contour Extraction → Alignment → PCA → Normal Shape Model

Contour features can include boundary coordinates, area, perimeter, aspect ratio, curvature and shape moments. A new component can be compared with the learned normal model; large deviations can indicate defects.

2. Proposed Classification Pipeline

Image Acquisition → Noise Reduction → Illumination Normalization → Object Segmentation → Contour/Feature Extraction → Feature Selection → SVM/GMM/CNN → Defective/Non-Defective.

SVM is suitable for small and medium datasets using engineered features. GMM can model the distribution of normal components and identify low-probability samples. CNNs can learn visual features directly when sufficient labeled images are available.

3. Handling Noise, Occlusion and Illumination

Noise: Gaussian/median filtering and morphological operations. Partial occlusion: contour completion, shape priors and augmented training examples. Illumination: normalization, histogram equalization, CLAHE and controlled industrial lighting.

Evaluation

Evaluate the system using accuracy, precision, recall, F1-score and a confusion matrix. Recall is particularly important because missing a defective component can have serious consequences.

Conclusion

A combination of PCA-based shape modeling, contour features and machine learning can provide an efficient defect-detection solution. CNNs become preferable when enough labeled images are available.

Question 2: Autonomous Vehicle Obstacle Recognition

Problem Understanding

The system must detect rare or unexpected obstacles on urban roads with limited labeled data while meeting embedded-system computational constraints.

1. GAN-Based Data Augmentation

A GAN contains a Generator that creates synthetic images and a Discriminator that distinguishes real from generated images. Synthetic examples can represent rare obstacles under different lighting, weather, backgrounds, scales, positions and viewpoints.

Benefits include increased training diversity and improved representation of rare classes. Generated images should be validated to avoid introducing unrealistic artifacts.

2. CNN vs Vision Transformer

Parameter	CNN	Vision Transformer
Feature extraction	Strong local features	Strong global-context modeling
Small objects	Effective with multi-scale features	Can be effective with suitable resolution
Data requirement	Generally more data-efficient	Often benefits from large-scale pretraining
Computation	Often lower for lightweight models	Can be higher
Embedded deployment	Easier to optimize	Requires careful optimization
Real-time use	Very suitable	Depends on model size and hardware

CNNs naturally capture local edges, textures and object parts. Vision Transformers use attention to capture relationships between image regions. For limited data and real-time embedded deployment, a lightweight CNN detector is generally the safer starting point.

3. Real-Time Inference Pipeline

Camera → Frame Acquisition → Image Resize/Normalization → Lightweight Object Detector → Feature Extraction → Bounding Box + Class Prediction → Confidence Filtering → Object Tracking → Vehicle Decision System.

Embedded optimization can use lightweight architectures, reduced input resolution, quantization, pruning, hardware acceleration and model compression. The design must balance accuracy, recall, FPS, memory and power consumption.

Conclusion

GANs can increase rare-obstacle training diversity, while a lightweight CNN detector provides an attractive balance between accuracy and computational efficiency for real-time embedded deployment.